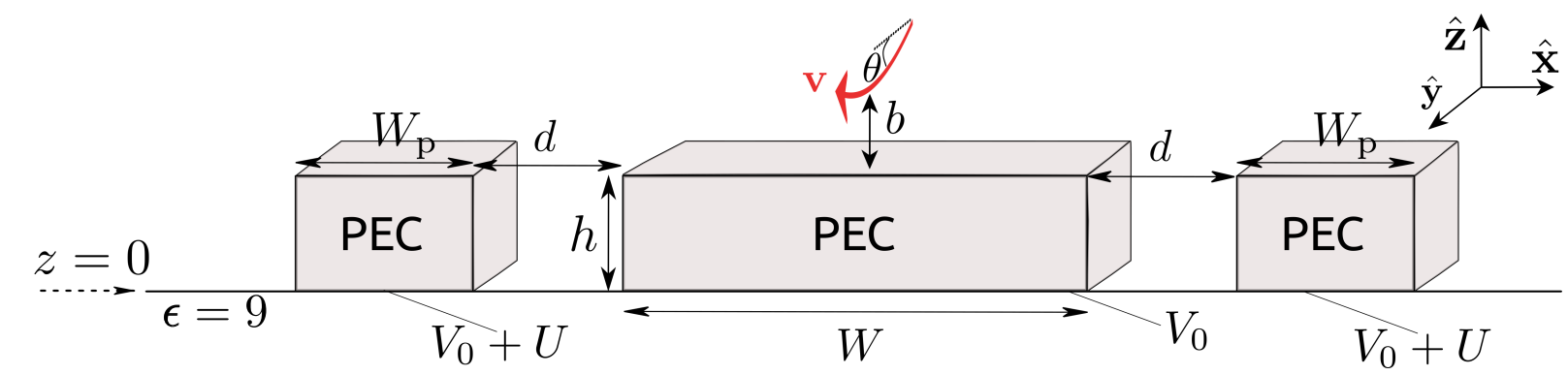


Electron-coupling to waveguide: code of the EELS

$$\Gamma_{\text{EELS}}(\omega) = 2 \int_{b_{\min}+h}^{\infty} \frac{v dz}{\sqrt{\left| c^2 \beta^2 \sin^2 \theta + \frac{2eV(x,z)}{m_e \gamma_e} \right|}} \underbrace{\frac{d\Gamma_{\text{EELS}}(\omega, x, z)}{dy}}_{\text{step 1}} \quad \text{integration in step 4}$$

$b_{\min}+h$
 $z_{\min}$
step 3b



1) run_EELS_rect_wg_along_z : obtain dEELS(z)/dy [1/eV*nm] for many energies

2) split_positive_eels.sh : split EELS in 1 file per 1 energy and retain only EELS >0

3) run_find_bmin_vs_V0_theta.sh :

a) Run e.out to create normalized potential data

g++ -o e.out e.cpp

files e.cpp and e.h

b) Run Python script to obtain bmin(V0,theta)
find_bmin_vs_V0_theta.py

solutions of this Eq:

$$0 = \frac{V(z_{\min})}{U} \frac{U}{V_0} + \frac{m_e c^2 \gamma_e}{2e} \frac{\beta^2 \sin^2 \theta}{V_0} \longrightarrow \theta, V_0 \text{ for a fixed } z_{\min}$$

4) EELS_integrated_over_z.h and EELS_integrated_over_z.cpp: save "energy | PIntegrated [1/eV]" for each solution V0, theta

g++ -o EELS_integrated_over_z.out EELS_integrated_over_z.cpp

5) plot_total_EELS_vs_theta.py: save "energy | EELS tot" for each solution V0, theta where EELS tot is the EELS integrated over z (data from step 4) and over the mode. Integration over the mode by fitting the data with a Lorentzian

fit_peaks.py